

Characterizing land use impact on multi-tracer displacement and soil structure



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SUMMARY

Leaching of solutes below the root zone has been identified as a main source of potential groundwater pollution. The occurrence of preferential flow paths in structured soils can enhance rapid leaching of solutes below the root zone. There is evidence that the actual land use can affect solute displacement by altering soil structure and the abundance of preferential flow paths. In the present study, a field experiment was conducted to assess the impacts of land use (grassland vs. no-till cropland) on profile-scale displacement of bromine (Br) and Brilliant Blue FCF. The objectives were (i) to study both solutes displacement patterns, (ii) to analyze the spatial variation and anisotropic variance structures of the solutes and controlling physical soil properties, and (iii) to analyze soil structure development as a result of the land use system and possible implications for solute displacement. Two ponding infiltration experiments with Potassiumbromide (KBr) and Brilliant Blue FCF were performed on a silt loam soil in Lexington, KY. A total of 30 mm multi-tracer solution was infiltrated on an area of 1.20×0.70 m. Eleven vertical profile sections (width: 1.10 m, depth: 0.80 m) were excavated in steps of 0.05 m and sampled. Dye stained areas were mapped based on digital image analysis. Small soil samples were taken for Br concentrations, soil texture, and volumetric soil water content at regular intervals along a vertical 0.10×0.10 m raster. Vane shear resistance was measured as a proxy for mechanical soil strength. X-ray fluorescence analysis was used to determine total Br contents and the relative SiO_2 signal intensity, the latter being used as proxy for soil particle size distribution. Although both experimental sites were under the same land use until some 10 years ago before the current land uses were established, solutes displacement differed between both land uses. The dye-stained patterns revealed a high proportion of non-equilibrium flow through vertically orientated macropores and a less permeable soil matrix at the grassland site. Continuous biological activity since transversion into grassland resulted in these macropores and the absence of any compaction in the subsoil. Large proportions of Br also infiltrated directly into the loose, densely rooted soil matrix close to the surface. Soil structure development at the no-till cropland site was mainly controlled by agricultural operations. The homogeneous Br distribution in the topsoil reflected a less dense soil matrix with a network of well-connected inter-aggregate pores. A residual plough pan restricted solute displacement to deeper soil layers or to groundwater bodies. Although ponding infiltration was applied in this study, the leaching risk for both applied solutes – Br and Brilliant Blue FCF – was rather small.

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1. Introduction

1.1. Solute transport in structured soil

Transport of solutes through agriculturally used soils has received increasing attention in recent years as solute displacement below the root zone has been identified as a main source

of potential groundwater pollution (Beven and Germann, 1982; Öhrström et al., 2004). Especially in structured soils, preferential flow paths and macropores can enhance rapid leaching of water and solutes below the root zone through bypassing the soil matrix under certain surface boundary conditions (Bouma and Dekker, 1978). Many experimental studies give evidence of these non-equilibrium water flow and solute transport phenomena (Gerke, 2006; Jarvis, 2007; Kodešová et al., 2011; Sander and Gerke, 2007; Wang and Zhang, 2011; Weiler and Naef, 2003). As small pores in the soil matrix are mainly predefined by soil texture, the

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occurrence and formation of structural pores are a result of complex interactions of climatic conditions – e.g., by formation of shrinking cracks – (Sander and Gerke, 2007), the activity of burrowing fauna such as earthworms (Cey and Rudolph, 2009), plant root growth and decay (Bodner et al., 2014; Nobles et al., 2010), soil type (Kodešová et al., 2011), and the actual land use (Bachmair et al., 2009). Based on these factors, Jarvis (2007) and Bormann and Klaassen (2008) stressed the need to better understand the impact of different land uses on soil structure and resulting pore geometry. As pointed out by Huisman et al. (2004), understanding land use impacts on water flow and solute transport is a prerequisite for better simulating possible impacts of land use change with respect to water resources management.

Numerous studies have investigated water flow at sites under one specific land use type (Bachmair et al., 2009). Moreover, most of the existing studies focussed on either differences in the hydraulic properties of the soil matrix or in the macropore system. For instance, Bormann and Klaassen (2008) compared physical and hydraulic soil properties under forest, grassland, and cropland. The study revealed considerable differences in bulk density, water retention, and hydraulic conductivity between the analyzed land uses. Bachmair et al. (2009) reported considerable differences in flow patterns of Brilliant Blue FCF dye under different land uses. The authors found a higher proportion of vertically oriented, continuous macropores at grassland sites compared to cropland. Schwen et al. (2012) analyzed Br[−] leaching in a silt loam soil under grassland and cropland and different irrigation characteristics. They reported that solute leaching was mainly controlled by the land use with a larger Br[−] displacement depth on the cropland. However, their experimental design permitted to assess the land use impact on soil structure and preferential flow pathways. Thus, there is a demand for comprehensive studies on the effects of land use on water flow and solute transport in the entire flow domain.

1.2. Tracers used to follow different flow domains

To observe flow processes of water and solutes in the field, different tracers have been used. Many studies used dyes such as Brilliant Blue FCF to study soil structure and identify preferential flow pathways (Kodešová et al., 2011; Nobles et al., 2010; Sander and Gerke, 2007; Wang and Zhang, 2011; Weiler and Naef, 2003). Brilliant Blue FCF is a food dye that is cheap, non-poisonous, and can be easily detected visually or by digital image analysis (Weiler, 2001). It is known to dissociate to a mono- or bivalent anion (depending on pH) and can also form ionic pairs with Ca²⁺ (Flury and Flühler, 1995) or adsorb to clay particles (Ketelsen and Meyer-Windel, 1999; Wang and Zhang, 2011). Thus, it has a restricted mobility in the soil pore system and can be used as proxy for the movement of larger organic molecules (e.g., pesticides) or to stain macropores and other preferential flow paths.

Contrarily, conservative inorganic tracers such as bromide (Br[−]) or iodine (I[−]) can be used for tracing water flow through the entire pore domain. These tracers have been mainly used to analyze solute flow and displacement processes in soil columns and at larger spatial scales (plot to field scale) (Flury and Flühler, 1995; Nobles et al., 2010; Öhrström et al., 2004; Schwen et al., 2012, 2013; Wang and Zhang, 2011). Due to its high solubility, small molecular size and low reactivity, these tracers can be used as proxy to observe the displacement of salts and nutrients (fertilizers) that are applied to agriculturally-used soils.

As outlined above, the tracers Brilliant Blue FCF and Br[−] are subject to different flow mechanisms (macropore-restricted flow vs. flow across the entire flow domain). Thus, the combination of both tracers in field solute experiments would allow a comprehensive description of solute displacement processes that are subject to treatments of interest (e.g., soil management or land use).

However, only few studies used tracer combinations to date (e.g., Bogner et al., 2008; Hangen et al., 2004; Nobles et al., 2010; Öhrström et al., 2004; Wang and Zhang, 2011). Nobles et al. (2010) analyzed solute leaching in caliche soils by using Br[−] and Brilliant Blue FCF. They found that Br[−] moved through the soil matrix in piston flow patterns, while Brilliant Blue FCF followed preferential flow paths. Moreover, Br[−] was less retarded and moved to greater vertical and horizontal distances. This was also reported by Bogner et al. (2008) for a sandy loam soil. Öhrström et al. (2004) found a retardation of Brilliant Blue FCF compared to Br[−] of 1.5 in a sandy loam soil. Wang and Zhang used a combination of I[−] and Brilliant Blue FCF to assess the influence of the ponding head and measurement scale on heterogeneous water flow in a structured clay soil. The authors concluded that water flow as indicated by the I[−] tracer contained more information on flow heterogeneity than the dye-stained macropores alone.

Another aspect that needs to be considered is the way how tracer solutions are applied to the soil. In some studies, tracers were applied by sprayers or sprinkling devices to the soil surface (e.g., Bachmair et al., 2009; Gerke and Köhne, 2004; Öhrström et al., 2004; Schwen et al., 2012; Weiler, 2001; Wendroth et al., 2011). Subsequently, the tracer was subject to irrigation treatments with different amounts or intensities to enhance infiltration into the soil. This application method simulates solute displacement as a result of natural rainfall mostly under unsaturated soil conditions. Other authors used ponding head infiltration of tracer solution within a framed area (e.g., Kodešová et al., 2011; Nobles et al., 2010; Sander and Gerke, 2007; Wang and Zhang, 2011). This application method represents rather extreme infiltration conditions. These may occur as a result of severe precipitation events that exceed the infiltration capacity of the soil. Ponding infiltration conditions can be used as a worst case scenario when leaching of nutrients or contaminants below the root zone is studied. Ponding infiltration of dye can also ensure a maximum staining degree of preferential flow paths and macropores that are open to the soil surface. The connection to the soil surface is required when dye tracers are used as a proxy to study soil structure and the macropore system (Kodešová et al., 2011). Wang and Zhang (2011) found that the distribution of small amounts of tracer solution (20–40 mm) was mainly restricted to the macropore network, while larger amounts (60–80 mm) inhibited a larger degree of matrix flow.

1.3. Detection methods for Br and major elements

The most common method to quantify total Bromine (Br) contents or Bromide (Br[−]) concentrations in soil samples is by using distilled water extracts and ion chromatography. However, this method comes along with a considerable sample preparation effort. To analyze a large sample number for Br contents, new methods are required. Lu and Wu (2003) presented an in-situ method, where Br[−] becomes visible in soil profile sections by a chemical reaction with silver ferrocyanide. Subsequently, the authors analyzed the profile sections by digital image analysis as applied for dye staining patterns. The method allows a detailed documentation of tracer distributions for 2-dimensional profile sections. However, the method also requires to analyze calibration samples by standard methods such as ion chromatography. Another disadvantage is that silver ferrocyanide is poisonous and its application may harm humans and the environment.

Abderrahim et al. (2011) detected total Br contents of dried soil samples by wavelength dispersive X-ray fluorescence (WD-XRF). The authors showed that the XRF method performs well for predicting Br contents that agree with Br[−] concentrations in water extracts. As solid samples are analyzed directly, this method requires a reduced sample preparation effort compared to ion

chromatographic techniques. Another advantage of XRF measurements would be that the signal intensities of major elements such as SiO_2 , Al_2O_3 , CaO , or MgO could be simultaneously detected. As reported by Acosta et al. (2011), SiO_2 and Al_2O_3 concentrations differ systematically between different soil particle size fractions due to their different mineralogy: The relative amount of quartz and feldspars is smallest in the clay fraction and increases with particle size. As the SiO_2 content is positively correlated with the particle size, and the Al_2O_3 content decreases with increasing particle size, both elements could be used as proxy for soil textural composition. However, to our knowledge this method has not been applied in combination with solute displacement experiments yet.

1.4. Flow-controlling physical soil properties

Besides the observation of water flow and solute transport in soils, knowledge of potentially controlling physical properties is required to improve our understanding of the underlying mechanisms. With respect to flow processes in the vadose zone, detailed information about the spatial variation of soil texture and structure is required. Several studies used the soil bulk density as an easily-measurable proxy for management-induced alterations in soil structure (e.g., Alvarez and Steinbach, 2009; da Silva et al., 1997; Schjønning and Thomsen, 2013). However, some authors questioned the usefulness of soil bulk density to represent flow-relevant soil structural characteristics (Håkansson and Lipiec, 2000; Horn, 2004; Schwen et al., 2011). By comparing different tillage systems, Horn (2004) demonstrated that the mechanical shear strength can document changes in soil structure as a result of changing soil management. The study suggested that the shear strength could be used as a proxy for soil structural variations.

1.5. Analyzing spatial variability of soil characteristics

A general challenge inherent in the measurement and description of solute leaching is the considerably high natural spatial variability of flow-controlling soil property functions and state variables (Netto et al., 1999). Moreover, most studies on spatial variations of soil hydrologic processes have been conducted in horizontal directions only (Schwen et al., 2014). As raised by Castrignanò et al. (2002), spatial variability in soils is characterized by variations in both the horizontal and vertical direction with a considerable anisotropy. This anisotropy is mainly due to pedogenetical soil horizons, as well as the occurrence of compacted soil layers (e.g., plough pan) or vertically-orientated macropores. The authors stressed the need to conduct more research on the 3-dimensional spatial variability of hydrological processes and relevant physical properties.

For quantifying the spatial variability of hydrological and soil-related observations, geostatistical methods have become useful tools (Nielsen and Wendroth, 2003). Variance analysis and computation of semi-variograms have been found to be adequate methods for analyzing spatial variations, also with respect to anisotropy (Remy et al., 2009). Moreover, the knowledge of variance structures allows the interpolation with 2- or 3-dimensional spatial models of a state-variable by kriging (Deutsch and Journel, 1997; Nielsen and Wendroth, 2003). In some cases, the data density of a quantity of interest is limited, whereas another related variable is known on a finer spatial or temporal scale. In that case, the covariance structure (cross-semivariogram) between these quantities can be calculated leading to improved interpolations of the sparsely known quantity by cokriging.

To analyze large data series, principal component analysis (PCA) – also known as empirical orthogonal functions (EOF) – has been applied in many hydrological and soil scientific studies (e.g., Crespo et al., 2011; Fischer et al., 2013; Vanderlinden et al.,

2012; Yuan et al., 2013). Within PCA, a data set is linearly transformed to a new coordinate system with the greatest and the second greatest variance being on the first and second principal components, respectively (Jolliffe, 2002). By this transformation, the internal structure of the data set can be identified and possibly correlated variables can be reduced into a smaller number of uncorrelated variables. Both statistical methods allow a comprehensive analysis of a set of soil-related data series with respect to its spatial variability and correlations among each other.

1.6. Study objectives

The motivation of the present study was to gain a deeper insight into solute transport mechanisms in a silt loam soil that was already studied by our working group (Wendroth et al., 2011; Schwen et al., 2012, 2013; Yang and Wendroth, 2014). Yang et al. (2013) stressed the need to assess the risk of deep leaching and groundwater contamination at this experimental site. The previous studies also indicated the demand to characterize macroporous structures as well as solute displacement by heterogeneous water flow. For the reasons pointed out above, a combination of Brilliant Blue FCF and KBr should be used in a field experiment. Since in fine-textured soils, dyes can penetrate into deeper soil only along the macropores (Wang and Zhang, 2011), the dye stained patterns should be used to describe the macropore network, while the Br distribution should reveal the heterogeneous flow patterns. To achieve a maximum in spatial resolution with a reasonable sample preparation effort, new observation and data exploration methods should be tested in the course of this study. The specific objectives of this study were (i) to quantify Br displacement applied to plots under two different land uses (grassland and cropland), (ii) to analyze the variability of the macropore system by dye staining patterns, (iii) to test the applicability of the mechanical shear strength as proxy for soil structural variations, (iv) and to assess correlations between static physical soil properties and the observed solute displacement.

2. Materials and methods

2.1. Field experiments

2.1.1. Experimental site and leaching experiment

The field leaching experiment was conducted during May 2013 at the University of Kentucky Agricultural Experiment Station Spindletop Farm in Lexington, KY (38°6'N 84°29'W). The soil at this site is a Maury silt loam, classified as a mixed, semiactive, mesic Typic Paleudalf according to the US Soil Taxonomy (Soil Survey Staff, 2010). The mean annual precipitation and temperature are 1170 mm and 12.8 °C, respectively. The experimental site consisted of two adjacent fields with different land use systems: One site was used as grassland with tall fescue (*Festuca arundinacea*), bluegrass (*Poa pratensis*) and red clover (*Trifolium pratense* L.) as the dominating species. The other site was used as an arable field cropped with no-till winter wheat (*Triticum aestivum*) (referred to as cropland). We acknowledge that both sites were part of one field that was entirely cropped with tobacco before being divided some 10 years ago.

Infiltration experiments were performed on both fields as described in the following. Four days prior to the tracer application, the soil water content was increased approximately to field capacity by drip irrigating a 2.0 × 2.0 m plot at both experimental sites. A rectangular steel frame with a length of 1.2 m and a width of 0.7 m was driven a few centimeters into the soil. In the cropland, the frame was orientated with the longer side being parallel to the wheat rows. Subsequently, the aboveground vegetation within

the framed area was cut with scissors directly above the soil surface and all plant residues and debris were removed with a vacuum cleaner. A total of 70 mm pre-mixed multi-tracer solution consisting of 10 g L⁻¹ KBr and 5 g L⁻¹ Brilliant Blue FCF was applied to the framed area maintaining a constant head of $h = 25$ mm (ponding infiltration). The Brilliant Blue FCF concentration was in the same range as reported from other studies (Flury and Flühler, 1995; Kodešová et al., 2011; Sander and Gerke, 2007). Schwen et al. (2012) used a KBr concentration of 39 g L⁻¹ for a field-scale solute leaching experiment in another area of the same experimental field. Nobles et al. (2010) used a KBr concentration of 30 g L⁻¹ in their multi-tracer experiment. However, since Brennan et al. (2012) used a KBr concentration of 4 g L⁻¹ in their study, we chose a concentration of 10 g L⁻¹ to be suitable for this experiment. The total infiltration depth of 70 mm was chosen to ensure that infiltration and solute displacement were not restricted to the macropore system (Wang and Zhang, 2011). The infiltration area was covered with a tarp until excavation to prevent evaporation.

2.1.2. Preparation of profile sections and photography

On the next day, vertical profile sections were excavated adopting the procedure of Wang and Zhang (2011). First, a profile pit (depth ≈ 0.9 m, width ≈ 1.4 m) in front of the longer side of the infiltration area (referred to as x-axis in the following) was dug out. A total of 11 vertical profile sections were then prepared in distances perpendicular to the x-axis (referred to as y-axis in the following) of $\Delta y = 0.05$ m starting 0.05 m behind the former position of the border frame (Fig. 1). A rectangular steel frame with

inner dimensions of $x = 1.10$ m and $z = 0.80$ m (where the z-axis denotes the depth) was fixed to the profile. The vertical and horizontal position of the frame was controlled by a level and steel poles that were driven vertically into the soil at both sides. The steel frame was equipped with horizontally and vertically orientated strings forming a regular raster with $\Delta x = \Delta y = 0.10$ m (Fig. 1). Since the soil surface was undulating in an amplitude of 0.03–0.04 m, the frame was installed at a constant depth throughout the sections ensuring that the uppermost horizontal string was always situated below the deepest surface position (Fig. 2a). This string was used as upper boundary for the subsequent analysis.

After preparation of the profiles, digital images were taken from a constant distance of 1.20 m with a resolution of 4608×3072 pixels. We used a Lumix DMC-G5X digital camera (Panasonic Corp.) that was mounted on a small tripod and equipped with a Lumix GX Vario PZ 14–42 mm lens and an external flash light. To ensure a continuous enlightenment and sun protection, the entire profile excavation was covered by a brown tent. A total of 11 images was taken at each experimental site (Figs. 1 and 2a).

2.1.3. Mechanical shear strength measurements

The undrained mechanical shear strength τ within each profile section was measured within a regular raster ($\Delta x = \Delta z = 0.10$ m, $\Delta y = 0.05$ m; $n_x = 10$, $n_y = 11$, $n_z = 7$) (Fig. 1). We used a vane shear test apparatus (16-T0174, Controls Inc., USA) with a vane dimension of $L = 0.04 \times D = 0.02$ m. The undrained shear strength τ (kPa) can be calculated from the torque M (Nm) and the vane dimensions L and D (mm) by the following equation (Farquhar et al., 2001):

$$\tau = \frac{M}{\pi D^2 \left(\frac{L}{2} + \frac{D}{6} \right)} \quad (1)$$

2.1.4. Soil core sampling procedure

Every other profile ($\Delta y = 0.10$ m, $n_y = 5$) was sampled for further laboratory measurements (see below). We used a small hand auger (inner diameter = 0.0187 m) to take undisturbed soil cores (length: 0.03 m, volume = 8.24×10^{-6} m³). Two adjacent cores were extracted within the same raster as the shear strength measurements ($\Delta x = \Delta z = 0.10$ m; $n_x = 10$, $n_z = 7$), but with an offset of $\Delta x = \Delta z = 0.05$ m. The applied sampling scheme is given in the detail picture at the bottom of Fig. 1.

2.2. Laboratory measurements

2.2.1. Determination of Br concentrations by XRF

One series of cores ($n_{\text{total}} = 350$ per land use) was used to determine the total Br content. We used a novel Br detection method with wavelength dispersive X-ray fluorescence (WD-XRF) as proposed by Abderrahim et al. (2011). The detection limit for Br was reported to be 1 mg (kg soil)⁻¹ (Abderrahim et al., 2011). All samples were air-dried and hand-ground. Cylindrical pellets (thickness: 0.01 m) were produced by putting 3 g of soil in crystalline boric acid (H₃BO₃), applying a pressure of 18 Tm for 20 s, and then allowing to decompress for 45 s. Analyses were performed with a S4 Pioneer WD-XRF spectrometer (Bruker/AXS GmbH, Karlsruhe, Germany) equipped with a Rh anticathode X-ray tube (1 kW, 50–20 kV, 20–50 mA), six analyzer crystals (GER, OVO-55, LiF 200, LiF 220, LiF 420 and PET), a flow-proportional counter for light element detection, and a scintillation counter for heavy elements.

In order to derive the Br concentration from the relative XRF signal intensity, calibration samples with 10, 100, 1000, and 10,000 mg Br (kg dry soil)⁻¹ were produced. We found a linear relationship between the XRF signal intensity and the Br content. The concentrations of the standards were validated using distilled

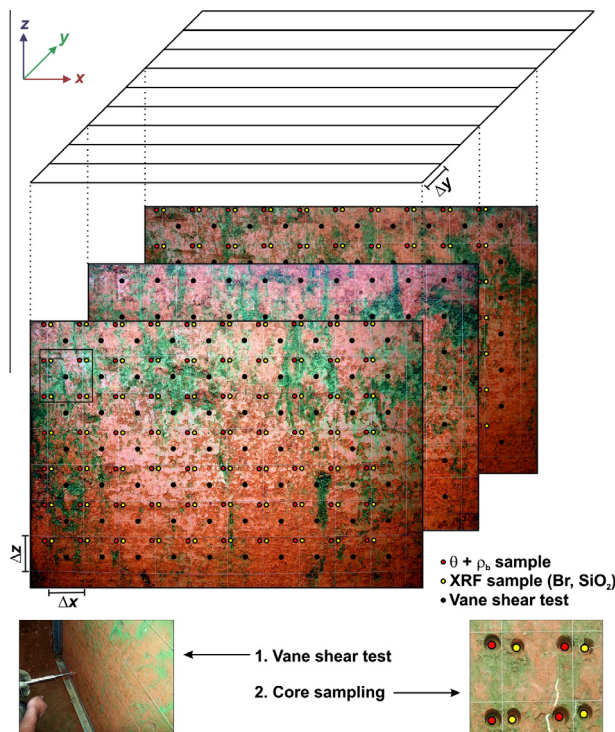


Fig. 1. Profile sections with sampling scheme. 11 Vertical profile sections were prepared with an offset of $\Delta y = 0.05$ m. All profiles were photographed and mechanical shear strength τ was measured with a vane shear test apparatus within a regular raster ($\Delta x = \Delta z = 0.10$ m; $n_x = 10$, $n_z = 7$) as indicated by the black dots. Every other profile ($\Delta y = 0.10$ m, $n_y = 5$) was sampled for volumetric soil water content θ , bulk density ρ_b (red dots), and XRF analysis (yellow dots) within the same raster ($\Delta x = \Delta z = 0.10$ m; $n_x = 10$, $n_z = 7$), but with an offset of $\Delta x = \Delta z = 0.05$ m from the shear strength measurement raster. Detail images of the shear tests and the sampling raster are given at the bottom. The definition of the cartesian axes (x, y, z) is also given. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

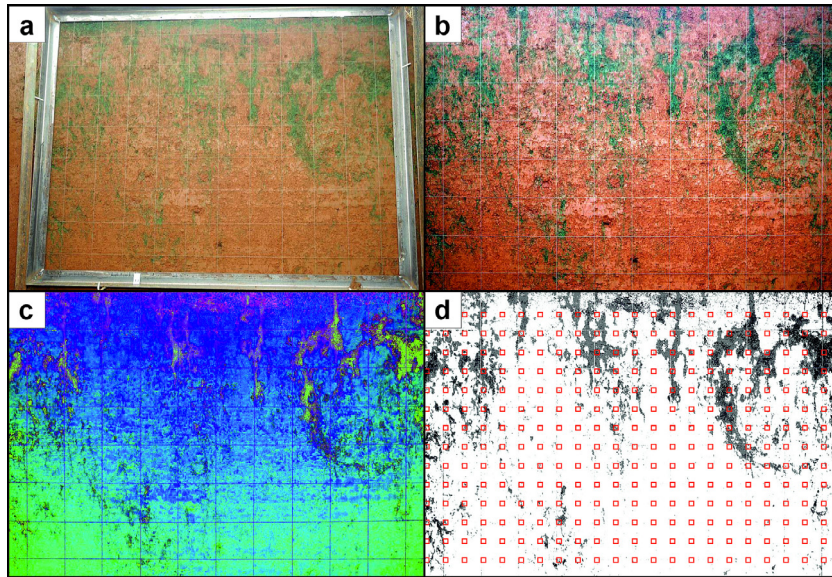


Fig. 2. Overview of the applied image processing steps. (a) Typical unaltered image of profile section. (b) After geometric and white balance correction. (c) After conversion to HSV colour space (miscolour illustration). (d) T values (Eq. (2)) as a measure for the dye staining degree. The red rectangles indicate the areas that were used to calculate mean T values prior to statistical analysis (area dimension: 0.01×0.01 m). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

water extracts and ion chromatography (Metrohm AG, Switzerland).

2.2.2. Qualitative soil texture measurements by XRF

2.2.2.1. Preliminary experiments. The XRF technique was also used to obtain a qualitative measure of the soil texture. Calibration samples were required to take advantage of the relation between particle size and contents of major elements (SiO_2 , Al_2O_3 , CaO and MgO). To estimate the range of particle sizes at the experimental site, we first used textural data from Schwen et al. (2013), who measured the particle size distribution to a depth of 1.00 m in the direct vicinity of the experimental sites of this study. When plotting those samples ($n = 480$) in a textural triangle, it became obvious that the sand content varied only little, whereas clay and silt contents varied considerably (Fig. 3). To produce calibration samples with predefined particle-size distributions, we separated the sand, silt, and clay fractions of a bulk soil sample that was taken at the experimental site. Calibration samples were then mixed along the observed texture gradient (Fig. 3). These calibration samples were used to determine the relative XRF signal intensities for the above mentioned major elements. The results confirmed that an increasing SiO_2 signal intensity correlated with a relative decrease of clay particles and increase of silt particles (intensities are given in Fig. 3).

2.2.2.2. Major element quantification by XRF. After clarifying methodological details, the XRF signal intensities of the major elements SiO_2 , Al_2O_3 , CaO , and MgO were measured simultaneously with the Br concentrations of all core samples. We acknowledge that this method only allows a qualitative estimation of the particle-size distribution. Therefore, textural information in the present study is discussed in terms of the SiO_2 XRF signal intensity.

2.2.3. Soil water content and porosity measurements

The second series of cores ($n_{\text{total}} = 350$ per land use) was used to determine the volumetric water content θ at the end of the infiltration experiment, the total porosity φ , and bulk density ρ_b thermogravimetrically (Fig. 1).

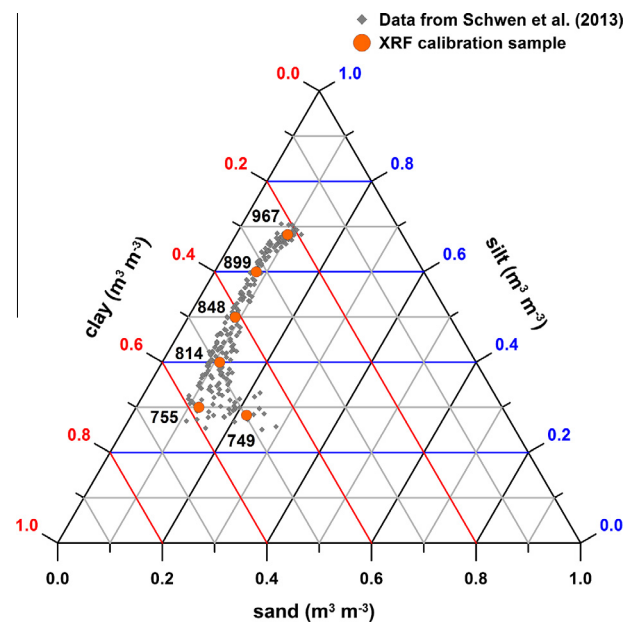


Fig. 3. Textural triangle with XRF signal intensities for SiO_2 . Measured textural data – i.e. the relative contents of sand, silt, and clay – at the experimental site were taken from Schwen et al. (2013). The orange dots show calibration samples with known texture for XRF calibration. The numbers next to the dots give the relative XRF signal intensity for SiO_2 . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2.3. Data processing

2.3.1. Digital image analysis

The pictures of the profile sections (raw image files) were processed to correct for geometric distortions and white balance using Silkipix Developer Studio (Version 5.0, Globell B.V.) and stored in tagged image file (TIFF) format (Fig. 2b). The subsequent image analysis was computed with Matlab (Version 2009R, The Mathworks Inc.). For the differentiation between dye-stained and non-stained areas (or the gradual staining degree), the images were

converted from the RGB (red–green–blue) to the HSV (hue–saturation–value) colour space as suggested by Weiler (2001) (Fig. 2c). The following equation was proven to separate the staining degree T independently from the underlying soil structure or colour (Weiler, 2001):

$$T = \left[\frac{H}{360\pi} (1 - V) \right] \cdot 100 \quad (2)$$

where H and V are the hue and values within the HSV colour space, respectively. Notice that – in contrast to other dye tracer studies (e.g., Kodešová et al., 2011; Kulli et al., 2003; Wang and Zhang, 2011; Weiler and Naef, 2003) – we did not use a threshold for T at this point of the data processing to classify the pixels as either stained (1) or non-stained (0). By keeping the full range of calculated T values, the information of small-scale gradual changes in the staining degree was preserved for further analysis (Fig. 2d). To facilitate further geostatistical analysis, the calculated T values were transformed into a 3-dimensional matrix with $\Delta x = \Delta y = \Delta z = 0.05$ m and $n_x = 22$, $n_y = 11$, $n_z = 15$. For this purpose, average T values were computed at each raster location from an area of 0.01×0.01 m (Fig. 2d).

2.3.2. Calculation of Br mass balance

To assess the degree of Br recovery in the analyzed soil compartment, the Br mass balance was calculated. The total amount of Br found in the soil was calculated from kriged values of ρ_b and Br in a 0.01 m-grid ($\Delta x = \Delta y = \Delta z = 0.01$ m). The total amount of Br being applied to the excavated area was 246.8 g at both sites.

2.3.3. Data preparation for statistical analysis

In order to facilitate the comparison of observation series with different magnitudes and ranges, we used the following transformation for all observations (Nielsen and Wendroth, 2003; Eq. 10.21):

$$y_{sc,i} = \frac{y_i - (\mu_y - 2\sigma_y)}{4\sigma_y} \quad (3)$$

Here, $y_{sc,i}$ are scaled values of the observations y_i , and μ_y and σ_y are the mean and standard deviation of the data series, respectively. This normalization results in values of $\mu_{sc,y} = 0.5$ and $\sigma_y = 0.25$ (Nielsen and Wendroth, 2003). Eq. (3) was applied to the observations of both land uses (with the same values for μ_y and σ_y). This normalization caused (i) variances of the same property observed at the two land uses and at different depths as well as (ii) variances of different properties observed at the same site to be independent of units and become directly comparable. All observation matrices were converted to the GSLIB file format (Deutsch and Journel, 1997) and were further analyzed with the Stanford Geostatistical Modeling Software (SGeMS; Remy et al., 2009).

2.4. Statistical analysis

2.4.1. Anisotropic variogram modelling

To facilitate 3-dimensional kriging, knowledge of anisotropic variance structures is required (Deutsch and Journel, 1997). In a first step, experimental semivariograms $\gamma(l)$ were calculated for all observations by (Isaaks and Srivastava, 1990):

$$\gamma(l) = \frac{1}{2N(l)} \sum_{i=1}^{N(l)} [A_i(x_i) - A_i(x_i + l)]^2 \quad (4)$$

Here, $A_i(x_i)$ denotes observation A_i at location x_i and $N(l)$ the number of observation pairs at the lag l . To assess the anisotropy, experimental semivariograms were computed in each axis direc-

tion (x, y, z). Subsequently, nested semivariogram models were fitted to the experimental semivariograms with the following general form (Remy et al., 2009):

$$\gamma(l) = c_0\gamma_0(l) + \sum_{i=1}^N c_i\gamma_i(l) \quad (5)$$

where γ_0 is a pure nugget effect and γ_i are either exponential or Gaussian variogram models, and $c_0, \dots, c_N$ are the variance contributions of each of the nested structures $\gamma_0, \dots, \gamma_N$. Exponential and Gaussian variogram models have the following general form (Nielsen and Wendroth, 2003):

$$\gamma(l) = c \left\{ 1 - \exp \left[- \left(\frac{l}{a} \right)^b \right] \right\} \quad (6)$$

where a and c denote the semivariogram range and sill, respectively, and the exponent b indicates the semivariogram model with $b = 1$ for the exponential and $b = 2$ for the Gaussian model type.

The above described semivariogram models (Eqs. 4–6) were given for isotropic conditions. These models have to be extended to account for possible anisotropies of variance structures. 3-Dimensional geometric anisotropy in SGeMS is described by an ellipsoid which is defined by three radii ($a_{\max}$, a_{med} , and $a_{\min}$) and three angles (α , β , and δ) in the following way: $a_{\max}$ denotes the range in the major direction of continuity. a_{med} and $a_{\min}$ are the ranges in the medium and minimum directions, respectively. The azimuth α describes the rotation of the ellipsoid about the z -axis. Values of $\alpha = 0$ and 90 denote an orientation along the y - and x -axis, respectively. The dip β describes the rotation of the ellipsoid about the x -axis. Values of $\beta = 0$ and 90 denote an orientation along the z -axis and the y - x -plane, respectively. The rake δ describes the ellipsoid rotation about the y -axis (set to 0 in this study). As the variability of observations (and its anisotropy) most often ranges over various scales, it can be described by a nested structure where each semivariogram model has its own anisotropy (Remy et al., 2009). The general model function with anisotropy and a nested structure of exponential or Gaussian semivariogram models is as follows:

$$\begin{aligned} \gamma(l_{\max}, l_{\text{med}}, l_{\min}) &= \sum_{i=1}^{N(l)} c_i \left\{ 1 - \exp \left[- \sqrt{\left(\frac{l_{\max,i}}{a_{\max,i}} \right)^2 + \left(\frac{l_{\text{med},i}}{a_{\text{med},i}} \right)^2 + \left(\frac{l_{\min,i}}{a_{\min,i}} \right)^2} \right]^{b_i} \right\} \\ &\quad \text{with } \alpha_i, \beta_i, \delta_i \end{aligned} \quad (7)$$

In this study, we used a nested semivariogram structure with $i = 2$ to account for the micro- and macro-structures, $\gamma_1(l)$ and $\gamma_2(l)$, respectively (Journel and Huijbregts, 1978). To describe the degree of anisotropy of each structure, we calculated the range ratios R_i :

$$R_i = \frac{a_{\min,i}}{a_{\max,i}} \quad (8)$$

To analyze anisotropical variance characteristics, we explored the experimental semivariograms along the main axes and the fitted semivariogram model functions. The estimation of semivariograms is limited by the number of pairs at the maximum lag distance $N(l)_{\max}$. In this study, observations along the y -axis were smallest and allowed to calculate variances $\gamma(l)$ up to lag distances of $l_{\max} = 0.50$ m and 0.40 m for T and all other observations, respectively (Fig. 8). With $N(l)_{\max} = 242$ and 70 for T and all other observations, respectively, a sufficient number of data pairs was available for semivariogram calculation.

As a result of the image analysis (Section 2.3.1), the dye staining degree was available in a dense raster ($\Delta x = \Delta y = \Delta z = 0.05$ m),

while Br concentrations were measured in a coarser raster ($\Delta x = \Delta y = \Delta z = 0.10$ m). As we expected the Br concentration to be influenced by the (stained) macropore network (Wang and Zhang, 2011), we calculated the cross-semivariogram $\Gamma(l)$ for the two data sets. Following Yates and Warrick (2002), $\Gamma(l)$ for two variables A_i and B_i , can be calculated by:

$$\Gamma(l) = \frac{1}{2N(l)} \sum_{i=1}^{N(l)} [A_i(x_i) - A_i(x_i + l)][B_i(x_i) - B_i(x_i + l)] \quad (9)$$

As for the semivariogram model (Eq. (4)), Eq. (7) can be adopted to model anisotropic cross-semivariograms.

2.4.2. 3-Dimensional kriging

After the fitting procedure, the derived anisotropic semivariogram models (Eq. (7)) were used to apply 3-dimensional ordinary kriging for all observations within SGeMS (Remy et al., 2009). Br concentrations were interpolated by cokriging under consideration of the variance structures of the Br concentration and the dye staining degree T and its cross-semivariogram (Eq. (9)). By these statistical interpolation procedures, values at unsampled locations within the 3-dimensional soil compartment for each observation were calculated (see e.g., Nielsen and Wendroth (2003) for details about the kriging method). As a result, observations that were taken at different spatial positions (Fig. 1) could be transferred to a common spatial grid for further analysis. We used a spatial grid discretization of $\Delta x = \Delta y = \Delta z = 0.01$ m with $n_x = 105$, $n_y = 50$, and $n_z = 70$ for the kriging procedure.

2.4.3. Principal component analysis (PCA)

To further evaluate the data set and reveal possible correlations, a principle component analysis (PCA; Jolliffe, 2002) was applied. We used the kriged data sets of both land uses ($n = 735,000$) for the observed quantities and the type of land use LU being coded as 0.25 and 0.75 for the grassland and cropland, respectively. We used the PCA procedure within SPSS (Ver. 15, IBM Corp.) with the Varimax rotation method. The adequacy of the data set for employing a PCA was given by a Kaiser–Meyer–Olkin value of 0.723. Components with Eigenvalues > 1.0 were considered as principal components only.

3. Results

In the following, the results of the solute leaching experiment are presented. We begin with a general description of the observed spatial distribution of the applied solutes and the controlling physical properties. Subsequently, the spatial behavior and variance structures of the analyzed data are discussed with respect to their anisotropy. Finally, the results of the PCA are used to analyze

correlations between the observed solute displacement and soil physical properties.

3.1. Solute displacement patterns

3.1.1. Br distribution and mass balance

The range of observed absolute Br contents was similar at both land uses with minimum and maximum values of 3.5 and 1320.9 mg Br (kg dry soil) $^{-1}$ in the grassland and 16.3 and 1419.9 mg Br (kg dry soil) $^{-1}$ in the cropland, respectively (Table 1). The arithmetic mean concentrations were 254.6 and 331.8 mg Br (kg dry soil) $^{-1}$ in the grassland and cropland, respectively. The total sums of Br found in the analyzed soil volumes were 204.2 and 228.7 g in the grassland and cropland, respectively, which resulted in a Br mass recovery of 83% and 93% in the grassland and cropland, respectively. The majority of Br was found in the topsoil (0–0.3 m depth) with the highest concentrations directly below the soil surface. Elevated concentrations were also found at some sampling positions close to the lateral borders of the excavated soil volume (Fig. 4a and b). This result suggests that the missing Br mass was very likely subject to lateral flow out of the analyzed soil volume. The lateral flow proportion was larger in the grassland (17%) than in the cropland (7%). The spatial Br distribution was rather homogeneous in the cropland, and the majority of the Br mass was found in 0.1–0.3 m soil depth (Fig. 4b, Video 2). Contrarily, the vertical Br displacement in the grassland was smaller with the highest concentrations directly below the soil surface (0–0.15 m depth) and showed a patchy distribution (Fig. 4a, Video 1).

3.1.2. Dye distribution and staining degree

The spatial Brilliant Blue FCF distribution correlated only moderately with the Br concentration patterns. This is in contradiction to findings of Öhrström et al. (2004) who reported a fairly good correlation between both tracers. The dye patterns at both sites varied on a smaller scale than Br and showed the characteristic small-scale patches (Figs. 4c and d and 5, Videos 3–4) as reported by Wang and Zhang (2011). Unscaled T values – a measure for the staining degree – were in the range of $3.95 \times 10^{-5} < T < 3.34 \times 10^{-2}$ and $6.93 \times 10^{-5} < T < 3.52 \times 10^{-2}$ in the grassland and cropland, respectively. Despite the similar T range, some noticeable differences between the two analyzed land uses were found. To illustrate these differences, we classified the T values in strongly stained ($T > 0.025$), moderately stained ($0.025 > T > 0.010$), and unstained ($T < 0.010$) and calculated the area fraction of each class as a function of profile depth (Fig. 6). In the grassland, most of the dye remained in the topsoil and the total (=moderately + strongly) stained area fraction decreased with

Table 1
Statistical characteristics of observations. Minimum, maximum, mean, and standard deviation of the measured properties at both analyzed land uses are given. This includes the bromine concentration Br, the dye staining degree T , the volumetric soil water content at the end of the infiltration experiment θ , the bulk density ρ_b , the mechanical shear strength τ , and the XRF signal intensities for the major elements SiO₂, Al₂O₃, MgO, and CaO.

	Br	T	θ	ρ_b	τ	SiO ₂	Al ₂ O ₃	MgO	CaO
	mg (kg soil) $^{-1}$	–	m ³ m ⁻³	Mg m ⁻³	kPa	XRF signal intensity			
<i>Grassland</i>									
min	3.5	3.96E–05	0.15	0.48	36.7	218.5	154.9	11.3	20.3
max	1320.9	3.34E–02	0.57	1.86	296.7	957.1	326.3	24.9	1045.7
arith. mean	254.6	5.50E–03	0.40	1.48	152.5	761.0	241.2	18.3	114.8
stand. dev.	236.4	6.04E–03	0.07	0.17	41.3	184.0	42.5	3.1	167.5
<i>Cropland</i>									
min	16.3	6.93E–05	0.25	0.96	38.0	666.1	176.4	13.6	13.3
max	1419.9	3.52E–02	0.50	1.92	252.0	982.5	322.5	22.9	28.9
arith. mean	331.8	4.99E–03	0.39	1.55	145.4	832.5	251.1	18.8	20.4
stand. dev.	267.5	5.74E–03	0.04	0.13	44.8	103.9	49.2	3.2	3.1

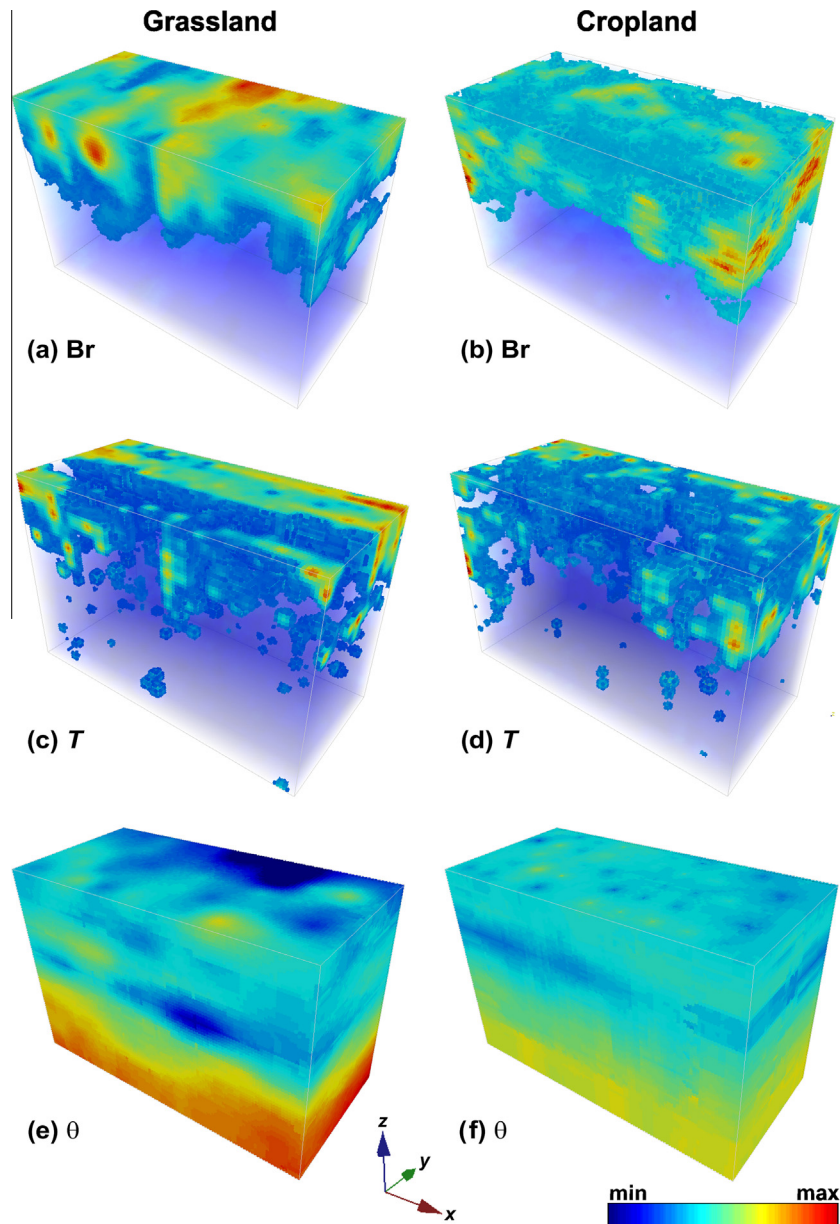


Fig. 4. Interpolated observations of the solute leaching experiment for both land uses (grassland and cropland). Scaled values for the bromine concentration Br (a, b), the degree of dye staining T (c, d), and the volumetric water content at the end of the experiment θ (e, f) are shown. Videos illustrating the 3-dimensional distribution of Br (a, b) and T (c, d) are available as online supplementary data (Videos 1–4). Parts with low values ($T_{\text{scaled}} < 0.65 \approx T_{\text{unscaled}} < 0.010$; $Br_{\text{scaled}} < 0.60 \approx 400 \text{ mg Br (kg dry soil)}^{-1}$) have been blanked in plots a–d for better visibility.

a linear trend from 0.56 below the soil surface to 0.05 in a depth of 0.50 m (0.25 m on the z -axis in Fig. 4a). Contrarily, the maximum total dye coverage in the cropland (0.50) was observed around 0.17 m below the soil surface (0.58 m on the z -axis in Fig. 4a), with a lower coverage close below the soil surface and in deeper areas. This indicates a considerable bypass of the topsoil matrix in the cropland. The fraction of the strongly stained area also differed between the land uses: In the uppermost 0.25 m, 17% of the total stained area was strongly stained in the grassland, whereas it was 29% in the same depth compartment in the cropland. It is noticeable that the dye pattern in the grassland showed a rather unstained area along the entire profile length (x -axis) at $0.10 \text{ m} < y < 0.30 \text{ m}$ (Fig. 4c, Video 3). This very likely reflects an increased abundance of preferential flow channels or macropores in this area.

3.2. Spatial distribution of flow-controlling properties

3.2.1. Soil water content and bulk density

Basic physical properties that indirectly control solute transport within the analyzed soil volumes were discussed by means of the bulk density ρ_b , the mechanical shear strength τ (Eq. (1)), and the XRF signal intensity for SiO_2 (Fig. 7), and also the soil water content at the end of the experiment. The volumetric soil water content θ ranged between 0.15 and $0.57 \text{ m}^3 \text{ m}^{-3}$ and 0.25 and $0.50 \text{ m}^3 \text{ m}^{-3}$ at the grassland and cropland, respectively. Both experimental sites were irrigated to field capacity four days prior to the infiltration experiment (i.e. five days prior to the sampling). Since evapotranspiration was minimized by cutting the plants and covering the soil surface with plastic tarps, the observed θ values are a result of vertical and lateral redistribution and the small-scale

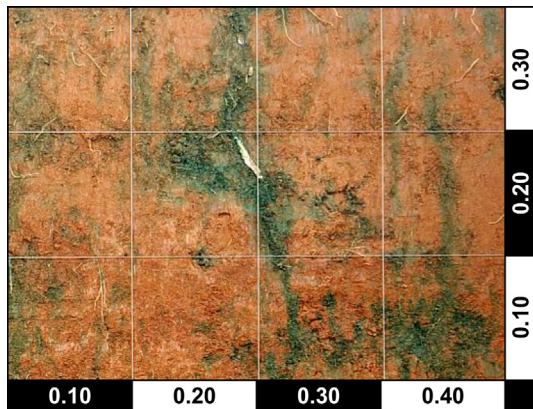


Fig. 5. Profile section with dye staining pattern at the grassland site. In the middle of the detail picture, a preferential flow channel is visible due to the dye staining. This area was intensively rooted and characterized by low mechanical shear strengths τ . Length dimensions are given in m.

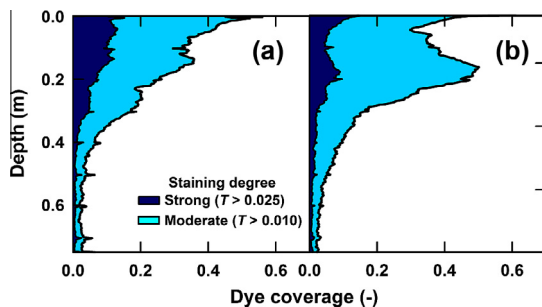


Fig. 6. Dye coverage as a function of soil depth for both land uses (a) grassland and (b) cropland. Area fractions with a strong and moderate staining degree were derived by classification of unscaled T values in each depth with $T > 0.025$ and $0.025 > T > 0.010$ for a strong and moderate staining degree, respectively. Area fractions with $T < 0.010$ were treated as unstained soil. Fractions are given as mean values from all profile sections ($n = 11$) at each land use.

water retention of the soil. Generally, the grassland showed wet and dry patches, especially close to the soil surface (Fig. 4e). This θ pattern corresponding with spatial variations of the bulk density ρ_b (Fig. 7a), with decreased water contents at denser areas. The highest θ values were found in the subsoil – i.e. below a depth of 0.50 m – at both land uses (Fig. 4e and f).

At the grass- and cropland, ρ_b was in ranges of 0.48–1.86 Mg m⁻³ and 0.96–1.92 Mg m⁻³, respectively. Spatial ρ_b variations in the grassland were more pronounced than in the cropland and no layering could be observed (Fig. 7a). The highest dry bulk densities in the cropland were found at depths below 0.30 m (Fig. 7b), with a sharp density decrease towards the topsoil. This finding can be attributed to the plough pan that still resided in the cropland while it appeared to be rather resolved in the grassland. The topsoil that had been ploughed until some 10 years ago was characterized by a rather homogeneous and low bulk density.

3.2.2. Mechanical shear strength

The mechanical shear strength τ ranged from 37 to 297 kPa and 38 to 256 kPa in the grassland and cropland, respectively. Despite a similar range and mean strength ($\tau_{\text{mean}} = 153$ and 145 kPa at the grassland and cropland, respectively), the spatial τ pattern differed considerably between the two land uses. In the grassland, irregular, small-scale variations and only a small increase with depth were found (Fig. 7c). Contrarily, a distinct τ increase between the topsoil (0–0.30 m) and the subsoil could be observed in the cropland (Fig. 7d). Within each horizon of the cropland, only small τ

variations were found. This finding agreed with the dry bulk density increase over depth that was mentioned earlier (Fig. 7b) and even better illustrated the effect of relatively loose topsoil and compacted subsoil presumably caused by cultivation of the cropland.

3.2.3. Major elements as a proxy for soil texture

XRF signal intensities were detected for the major elements SiO₂, Al₂O₃, MgO, and CaO. We found strong negative correlations between the SiO₂ and Al₂O₃ or MgO signal intensities ($R^2 = 0.98$). Thus, the latter properties did not contain additional information and were not included in the results.

The SiO₂ signal intensity ranged from 219 to 957 and from 666 to 982 in the grassland and cropland, respectively. The range observed in the cropland agreed with the calibration range (Fig. 3). In the grassland, however, some samples resulted in SiO₂ signal intensities below 650. To understand this behavior, we also measured the CaO signal intensity, which ranged from 20 to 1046 and from 13 to 29 in the grassland and cropland, respectively. The considerably high CaO intensities in the grassland were only found at the bottom of the analyzed soil (depth > 0.65 m). Here, we also detected the low SiO₂ intensities. This finding can be explained by the occurrence of weathered carbonatic bedrock debris in the subsoil at this site, that was noticed during the profile excavation. Besides these differences in the subsoil of both profiles, the spatial variations of the SiO₂ signal intensities were rather similar in both land uses. The highest intensities were found in the topsoil (0–0.30 m depth), whereas relatively low intensities were detected in the subsoil. The transition zone between the silt-dominated topsoil and the clay-dominated subsoil was rather sharp in both profiles.

3.3. Anisotropic variance structures of observations

The above described spatial distributions of the solutes (Br and Brilliant Blue FCF) and physical soil properties already indicated that considerable spatial variations for most of the properties were observed. Obviously, variations with depth (along the z -axis) were more pronounced than in horizontal directions (along the x - and y -axis). In the following section, spatial variations of the observations were analyzed by the fitted anisotropic semivariogram models.

3.3.1. Br concentrations and dye staining degree

Semivariances of the Br contents and dye staining degree T showed similar characteristics at both land uses (Fig. 8a–d). As variances along the x - and y -axis differed only little, horizontal variations were found to be almost isotropic. Only the horizontal variations of the T values at the grassland showed distinct differences between the x - and y -direction (Fig. 8c). Here, variations along the y -axis were larger with unidirectional sills of $c_x \approx 0.038$ and $c_y \approx 0.050$. This finding can be explained by the unstained area at $0.10 \text{ m} < y < 0.30 \text{ m}$ that was mentioned earlier (Fig. 4c, Video 3). Interestingly, we found only negligible variance differences between the x - and y -axis in the cropland (Fig. 8b and d). As the underlying observations were taken parallel and perpendicular to the crop rows along the x - and y -axis, respectively, this finding indicated that the cropping direction and sampling position (row or interrow) did hardly affect the solute displacement for both Br and Brilliant Blue FCF. Compared to horizontal variances, variances along the z -axis were much larger for both solutes at both land uses (Fig. 8a–d). In the vertical direction, semivariances increased with lag distance l unbounded and almost linear. This could be explained by the fact that the solute concentrations decreased with depth (Fig. 4a–d, Videos 1–4). When fitting semivariogram models to the experimental semivariograms, this pronounced anisotropy was reflected by small anisotropy ratios of the macro-variance

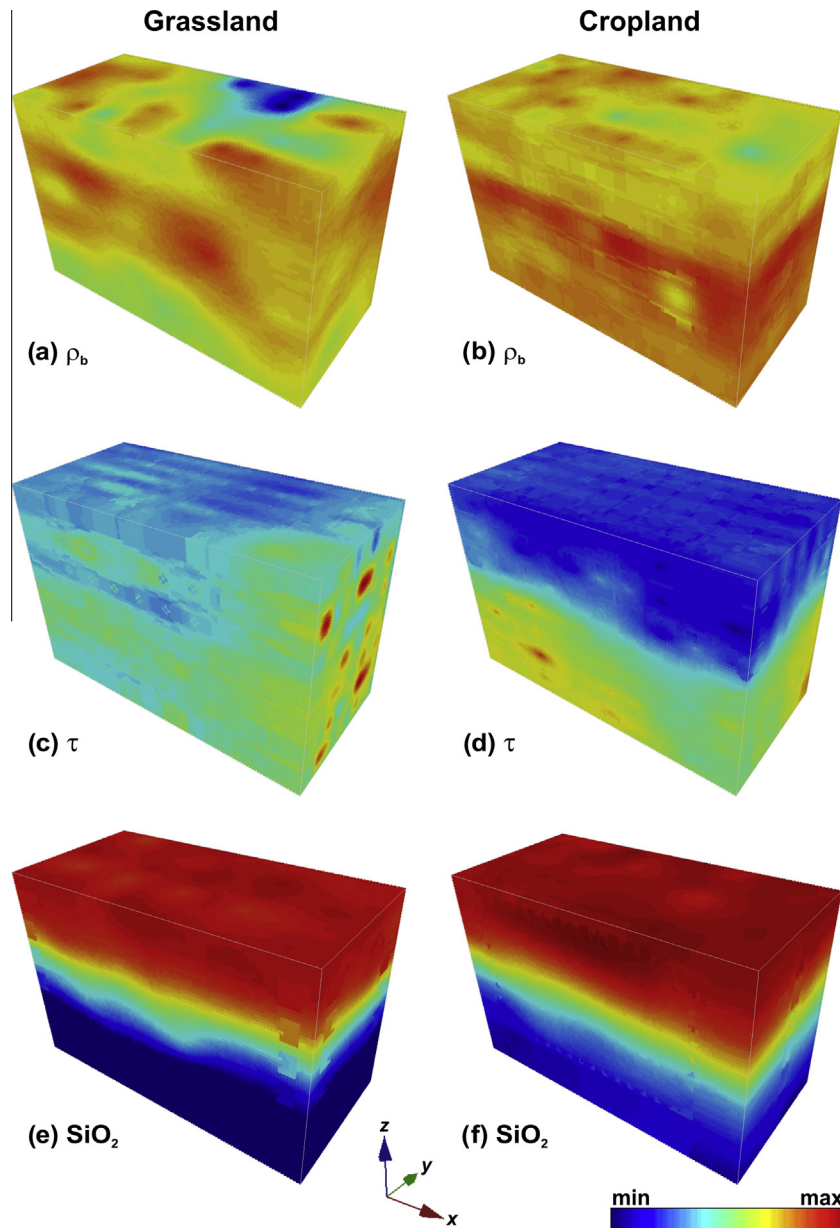


Fig. 7. Interpolated observations of physical soil properties for both land uses (grassland and cropland). Scaled values for the bulk density ρ_b (a, b), the mechanical shear strength τ (c, d), and the relative silicone XRF signal intensity SiO_2 (e, f) are shown.

structures ($0.14 < R_{\text{macro}} < 0.29$; Table 2). When comparing the variances for the Br and T values, we found a higher small-scale variation for the T values than for Br at both land uses (Fig. 8a–d). This finding was reflected by short ranges for the T value micro-variance structures in all directions ($a_{\text{max},1} = a_{\text{med},1} = a_{\text{min},1} = 0.05$ m and 0.06 m; Table 2). We acknowledge that this variance structure could also be described by a nugget effect ($c_0 \approx 0.035$ – 0.040). However, we found that the Gaussian model used for small-scale variances better fitted the dye patterns as reflected by the profile images. The small-scale T value variances manifested the patchy dye tracer distribution (Fig. 4c and d, Videos 3, 4) and the short displacement length of Brilliant Blue FCF around macropores (Fig. 5). Contrarily, the smaller micro-variance structures of Br indicated a longer displacement length into denser and stronger aggregated soil areas that led to a more homogeneous tracer distribution (Fig. 4a and b, Videos 1, 2).

3.3.2. Soil water content and bulk density

Horizontal variances in the soil water content θ in the cropland were isotropic and characterized by a pure nugget effect ($c_{0,xy} \approx 0.015$; Fig. 8f). In the grassland, larger variances for the y -direction were found at lag distances $l > 0.20$ m (Fig. 8e). This finding agreed with the higher θ values that were found in the rear area of the analyzed soil (Fig. 4e). As the horizontal variance structure of the bulk density ρ_b showed a similar anisotropy (Fig. 8g), it could be concluded that lateral changes in the soil water retention were positively correlated with changes in the soil bulk density. Vertical θ variances were – like variances of the solutes Br and Brilliant Blue FCF – unbounded and much larger than in horizontal directions. Notice, the unbounded semi-variance behaviour was not caused by a trend in magnitude across depth, but was indeed an increasing variance of the normalized data. This was reflected by anisotropy ratios of

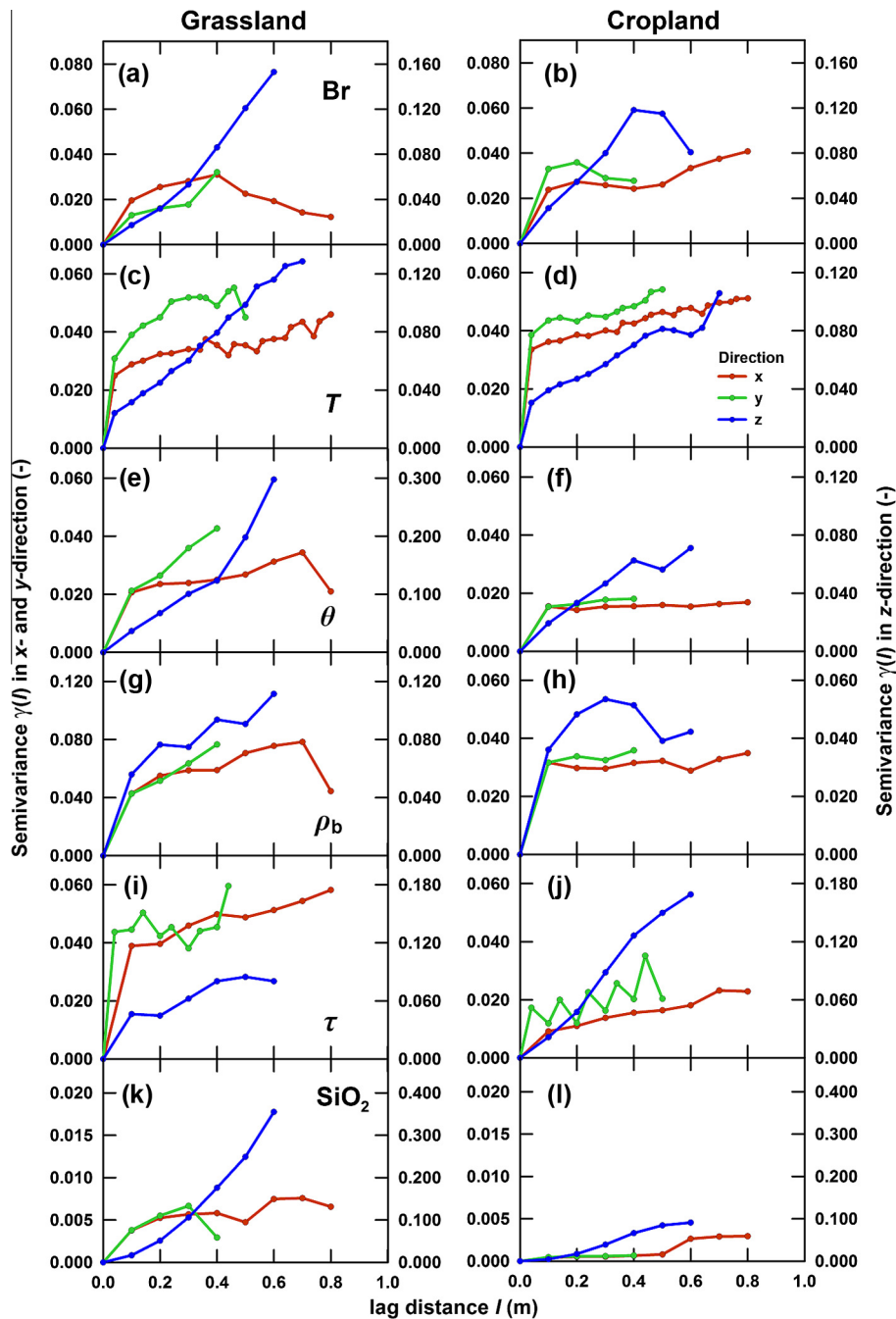


Fig. 8. Experimental anisotropic semi-variograms $\gamma(l)$ (Eq. (4)) for all observed properties at both experimental sites (grassland and cropland). Variograms of normalized data are given for the bromine concentration Br (a, b), the degree of dye staining T (Eq. (2); c, d), the volumetric water content at the end of the experiment θ (e, f), the bulk density ρ_b (g, h), the mechanical shear strength τ (Eq. (2); i, j), and the relative silicone XRF signal intensity SiO_2 (k, l). Due to the pronounced anisotropy, γ values for the x - and y directions are plotted on the left y-axis, whereas γ values for the z -direction are plotted on the right y-axis. All data were normalized by Eq. (3) prior to statistical analysis.

$R_{\text{micro}} = 0.58$ and 0.11 and $R_{\text{macro}} = 0.13$ and 0.03 at the grassland and cropland, respectively (Table 2). Variances of ρ_b showed similar characteristics (Fig. 8g and h). However, it should be noticed that ρ_b variances in the grassland were in the same order of magnitude in all directions (Fig. 8g), and were thus the only quantity that showed an isotropic variance pattern on the micro- and macro-scale. The dry bulk density at the cropland was characterized by a maximum in a depth of 0.30 – 0.40 m (plough pan) and lower densities below and above (Fig. 7b). Therefore, the semi-variogram along the z -axis showed a maximum at $l \approx 0.30$ m and decreased for larger lag distances (Fig. 8h).

3.3.3. Mechanical shear strength

Variances of the mechanical shear strength τ differed considerably between the land uses. In the grassland, small-scale variances in all directions were characterized by a sill of $c_1 = 0.040$ (Table 2), reflecting a considerable small-scale variability. However, variances in the vertical direction showed a relatively smaller variance at the micro-scale and larger variances at the macro-scale, as reflected by $R_{\text{micro}} = 0.38$ and $R_{\text{macro}} = 0.33$, respectively (Table 2). In the cropland, small-scale variances in all directions were smaller ($c_1 = 0.010$; Table 2). Also, variances in horizontal directions (x - and y -direction) were smaller on the macro-scale, indicating a rather

Table 2

Anisotropic variogram models. The semivariogram models $\gamma(l)$ for all measured quantities and the cross-semivariogram models $\Gamma(l)$ for the Br concentration and the dye staining degree T are given. The following parameters are given: the sill c_i , the model type exponent b_i (with $b_i = 1$ for an exponential and $b_i = 2$ for a Gaussian model), the anisotropic ranges $a_{\max,i}$, $a_{\text{med},i}$, and $a_{\min,i}$, and the ellipsoid-defining angles azimuth α_i and dip β_i . We acknowledge that we set all rake angles $\delta_i = 0$. Nested models with $i = 2$ were used where $\gamma_1(l)$ and $\gamma_2(l)$ describe the micro- and macro-structures, respectively. R_1 and R_2 denote the anisotropy ratios (Eq. (8)) of the micro- and macro-variance structures, respectively.

	c_1	b_1	$a_{\max,1}$	$a_{\text{med},1}$	$a_{\min,1}$	α_1	β_1	R_{micro}	c_2	b_2	$a_{\max,2}$	$a_{\text{med},2}$	$a_{\min,2}$	α_2	β_2	R_{macro}
	–	–	m			–	–	–	–	–	m			–	–	–
Grassland																
Br	0.030	1	0.58	0.24	0.24	0	0	0.41	0.70	2	7.87	7.72	2.32	90	0	0.29
T	0.025	2	0.05	0.05	0.05	90	0	1.00	0.13	1	15.71	5.60	1.72	90	0	0.11
θ	0.025	1	0.19	0.19	0.11	90	0	0.58	0.70	2	11.89	4.32	1.54	90	0	0.13
ρ_b	0.040	1	0.18	0.18	0.14	90	0	0.78	0.10	1	3.87	2.80	1.40	90	0	0.36
τ	0.040	2	0.16	0.06	0.06	90	0	0.38	0.09	2	3.80	3.80	1.24	90	0	0.33
SiO ₂	0.0006	1	0.33	0.33	0.33	90	0	1.00	0.45	2	30.88	11.12	0.93	90	0	0.03
Br/ T	0.01	1	0.80	0.29	0.07	90	0	0.09	0.06	2	2.52	2.52	0.51	0	0	0.20
Cropland																
Br	0.028	1	0.20	0.20	0.20	00	0	1.00	0.09	2	4.40	3.09	0.60	90	0	0.14
T	0.035	2	0.06	0.06	0.06	90	0	1.00	0.07	1	7.43	3.66	1.61	90	0	0.22
θ	0.015	1	0.80	0.09	0.09	0	90	0.11	0.05	1	21.52	17.22	0.65	90	0	0.03
ρ_b	0.030	1	0.13	0.13	0.13	90	0	1.00	0.03	1	10.76	4.84	0.65	90	0	0.06
τ	0.010	1	0.15	0.07	0.07	90	0	0.47	0.13	2	3.66	1.83	0.54	90	0	0.15
SiO ₂	0.0006	1	0.22	0.22	0.22	90	0	1.00	0.10	2	15.44	15.44	0.76	90	0	0.05
Br/ T	0.012	1	0.39	0.20	0.20	90	0	0.51	0.08	2	8.82	5.38	0.62	90	0	0.07

homogeneous strength pattern. However, it should be noticed that variances along the y-axis were characterized by distinct undulations (Fig. 8j). As the y-axis represents the sampling direction perpendicular to the cropping rows, we refer these undulations to alterations in the mechanical soil strength by the wheat crops (row or interrow sampling position). Vertical semivariograms in the cropland increased linearly and were unbound (Fig. 8j) and larger compared to the grassland.

3.3.4. SiO₂ signal intensity

Variances of the SiO₂ signal intensity were characterized by a pronounced anisotropy between horizontal and vertical directions as reflected by R_{macro} values of 0.03 and 0.05 for the grassland and cropland, respectively (Table 2). Compared to the cropland, horizontal and vertical variances were larger in the grassland (Fig. 8k and l). The considerable vertical variances could be explained by the textural gradient and the occurrence of calcareous bedrock debris with low SiO₂ content at this site (Fig. 7e). As mentioned earlier, both experimental sites were part of one larger field cropped with tobacco until some 10 years ago. At that time, soil management operations (tillage) should have mixed and homogenized soil particles and thus minimized horizontal variations. As this effect is still reflected by the very small horizontal variances in the cultivated cropland (Fig. 8l), the larger horizontal variances in the grassland might be a result of biologically-induced perturbations resulting in small-scale heterogeneity.

Table 3

Correlation matrix of observations. The dye staining degree T , the bromine concentration Br, the bulk density ρ_b , the relative silicone XRF signal intensity SiO₂, the mechanical shear strength τ , the volumetric water content θ , and the land use LU were evaluated.

	T	Br	ρ_b	SiO ₂	τ	θ	LU
T	1.000	0.626	−0.169	0.517	−0.510	−0.524	−0.059
Br	0.626	1.000	−0.123	0.749	−0.740	−0.708	0.224
ρ_b	−0.169	−0.123	1.000	0.274	0.233	−0.110	0.413
SiO ₂	0.517	0.749	0.274	1.000	−0.596	−0.906	0.272
τ	−0.510	−0.740	0.233	−0.596	1.000	0.581	−0.102
θ	−0.524	−0.708	−0.110	−0.906	0.581	1.000	−0.136
LU	−0.059	0.224	0.413	0.272	−0.102	−0.136	1.000

3.4. Correlations among observations and PCA

In this final section, we analyzed spatial relationships between solute displacement and physical soil properties. First, a simple correlation matrix was calculated in the course of the PCA (Table 3). The correlation between both solutes for the entire data set (both land uses) was $\text{Br}/T = 0.626$. If both solutes, Br and Brilliant Blue FCF, would have been subject to the same infiltration process, the slope of the linear regression would be ideally 1 and the spread reflected by an R^2 value close to 1. A higher deviation from this 1:1 line indicates a stronger degree of flow separation of bromine and dye. In our study, the linear regression coefficients were small in both grassland (0.531) and cropland (0.427). This indicated a high degree of flow separation between both tracers regardless of land use (Fig. 9). Moreover, both solutes showed a large spread around the correlation line as reflected by R^2 values of 0.373 and 0.233 at the grassland and cropland, respectively (Fig. 9). This result supported the finding that Br concentrations differed considerably from the dye patterns.

As we found negative correlations between τ and θ with all other variables (Table 3), these data series were multiplied by -1 prior to PCA. As a result of the PCA, two principal components with Eigenvalues of 3.64 and 1.58 were identified that explained 51.5% and 23.1% of the total variance, respectively (Fig. 10). The Varimax rotation method resulted in a rather small rotation of the coordi-

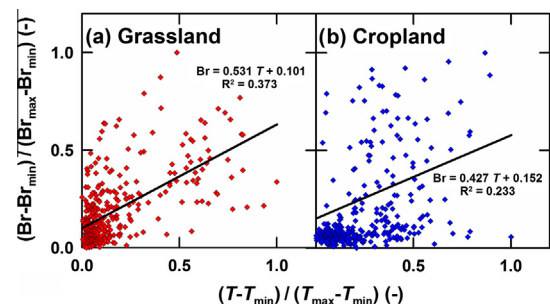


Fig. 9. Correlation between bromine concentration Br and dye staining degree T observed in (a) the grassland site and (b) the cropland site. The data were normalized to the maximum value Br_{max} and T_{max} , respectively. Linear regressions and the corresponding equation are also given.

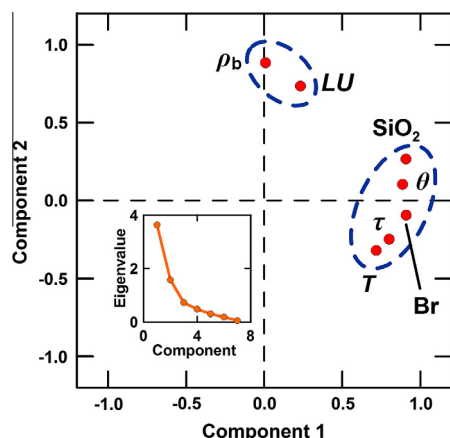


Fig. 10. Results of the principal component analysis. Two principal components with Eigenvalues >1.0 were identified (small scree plot) where components 1 and 2 explained 51.5 and 23.1% of the variance, respectively. The component plot shows the loadings of both components for the bromine concentration Br, the degree of dye staining T , the volumetric water content θ , the bulk density ρ_b , the mechanical shear strength τ , the relative silicone XRF signal intensity SiO_2 , and the land use LU .

nate system (rotation angle: 0.99°). Component 1 was highly charged by Br (0.912), $-\theta$ (0.862), SiO_2 (0.861), $-\tau$ (0.827), and T (0.755), whereas component 2 was highest charged by ρ_b (0.878) and the land use LU (0.760). The PCA revealed the existence of two aggregated groups of variables with similar component loadings (Fig. 10). The PCA and the correlation matrix indicated that variations between the two land uses (grassland vs. cropland) were mainly manifested in ρ_b variations (reflected by similar component loadings in Fig. 10). On the other hand, variations of the observed solute displacement (Br and T) were strongly related to variations in the mechanical shear strength τ , and to a slightly smaller extent to variations in the soil water retention θ , which in turn was correlated to variations in the soil texture (relative silicone XRF signal intensity SiO_2). The importance of soil texture and differently textured soil horizons for water flow and solute transport patterns was also pointed out by Kulli et al. (2003).

4. Discussion

4.1. Advantages and limitations of methodology

4.1.1. Ponding head infiltration

In this study, both tracers were applied under ponding head conditions. Compared to the application by sprinkling systems or sprayers, it represents rather extreme infiltration conditions that occur only when the precipitation exceeds the infiltration capacity of the soil. We acknowledge that the results of this study cannot be directly transferred to lower, more frequently occurring precipitation conditions. It can be expected that especially the degree of non-equilibrium flow through the macropore network might be much smaller when sprinkling would be applied. This assumption is confirmed by comparing our results with the study of Schwen et al. (2012). In that study, Br^- displacement as a result of different irrigation amounts and intensities was observed at the same experimental site. The irrigation treatments were opposed by a sprinkling system and the mean Br^- displacement depth, expressed as Br^- center of mass, was 0.08 and 0.11 m at the grassland and cropland, respectively. These values were considerably smaller than the mean displacement depths in this study (Fig. 11). Thus, the Br displacement of this study represents a worst case scenario that leads to a maximum displacement.

With respect to Brilliant Blue FCF, the ponding infiltration method ensured a maximum staining degree of macropores and preferential flow paths. As for the Br, the staining degree might be much smaller when the tracer is applied by sprinkling. This assumption is confirmed by the study of Bachmair et al. (2009), who analyzed the infiltration depth and staining pattern of dye subject to different irrigation amounts. The authors observed an increased dye staining depth with increasing irrigation amounts under grassland and cropland. In our study, the dye staining patterns were basically used as a proxy for macropores and other preferential flow structures that are accessible from the soil surface. We acknowledge that there might be more macropores in the sub-soil that were not identified by this method – especially below the compacted plough pan at the cropland. However with respect to the Br displacement, only macropores connected to the soil surface that contribute to preferential flow were identified.

4.1.2. Br and major element detection by XRF

This study was the first that tested the XRF Br detection method presented by Abderrahim et al. (2011) in a field solute displacement experiment. Our results confirmed a good applicability of the XRF method. The range of Br concentrations (Table 1) could be fully covered by the applied XRF detection method. We found a good agreement between Br concentrations obtained by XRF and Br^- concentrations determined by ion chromatography in distilled water extracts. The mass balance revealed a high degree of Br recovery in the analyzed soil volumes (cp. Section 3.1.1). As for ion chromatography, calibration samples with known tracer concentration were required. Besides the low effort for sample preparation, the XRF method also allowed to detect signal intensities for the major elements. The signal intensities were either positively (SiO_2) or negatively (Al_2O_3 and MgO) correlated with the particle size distribution and thus confirmed the findings of Acosta et al. (2011). We conclude that signal intensities for one of these elements were sufficient to derive a qualitative measure of the particle size distribution. Calibration for local site conditions remains in place until more experience has been gained.

However, we also found a limitation of this method: As the classical texture triangle with relative contents of sand, silt and clay is a 2-dimensional space (Fig. 3), it is not possible to derive individual “positions” (i.e., exact fractions of sand, silt and clay) from one of these quantities (e.g., SiO_2). Therefore, this method can be only used as a qualitative measure of a mean sample particle size distribution.

The CaO signal intensities were positively correlated with the inorganic carbonate content of the soil and were useful to identify the abundance of limestone debris in the subsoil. As a conclusion, the detection of major element signal intensities by XRF gave valuable information about soil texture at individual locations with a strongly reduced sampling effort compared to the classical sieving and sedimentation method.

4.1.3. Representativity of point measurements and experimental sites

Within this study, a new sampling scheme was tested where all observations except the dye staining pattern were determined as point measurements within regular $0.10 \times 0.10 \times 0.10$ m rasters. It should be noted that core sampling and vane shear tests are destructive methods that cannot be applied at the same position. Therefore, there was an offset between the sampling positions for Br concentrations, soil water content/bulk density, and mechanical shear strength. One might argue that this sampling scheme resulted in a limited representativity since (i) the size of the core samples was rather small, and (ii) variations that occurred on a smaller scale than the sampling raster might not be covered. However, the ranges of the semivariograms were larger than the sampling distances (Table 2). From the studies of Nobles et al.

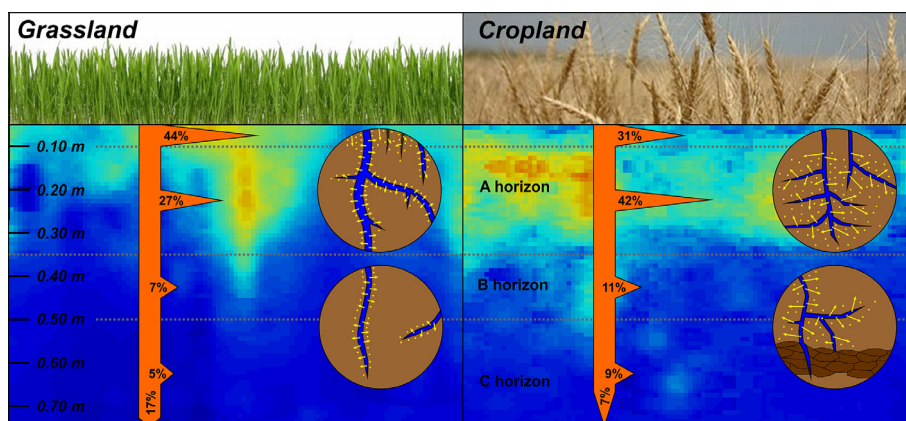


Fig. 11. Illustration of the Br displacement and major flow processes at the grassland (left) and cropland (right). The background picture shows the Br concentration of a representative profile section for each land use. The main diagnostic horizons (A, B, and C horizon) are depicted. The orange bar indicates the relative Br displacement proportion in each horizon as well as the proportions that remained directly below the soil surface and that were subject to deep or lateral drainage (see Section 3.1.1 for details on the Br mass balance). The circled illustrations reflect the major flow processes, where the blue areas indicate dye-stained macropores and preferential flow paths and the yellow arrows and dots the displacement direction and increased concentrations of Br, respectively. The ploughpan at the cropland is illustrated by a layer of dense soil aggregates (dark brown), while plant roots are depicted at the top of the grassland soil. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(2010) and Wang and Zhang (2011), it could be also expected that the Br concentrations depended on the distance from preferential flow paths. In both studies, detailed tracer distributions were made visible by chemical application on profile sections (Lu and Wu, 2003). This dependence might not be covered by the regular sampling scheme of our study.

However, the regular sampling raster resulted in a large number of sampling positions that were either within preferential flow areas or in the soil matrix with different distances from preferential flow paths. This was confirmed by the large ranges of magnitudes found for Br concentrations and all other observations (Table 1). We also acknowledge that the tested sampling scheme delivered a higher data density compared to auger sampling campaigns from the soil surface. The latter is often used to measure field-scale solute distributions (e.g., Schwen et al., 2012). It should be also noted that we accounted for the influence of the macropore network on Br displacement by cokriging. By this operation, the resulting Br concentrations incorporated detailed information about the dye staining pattern in a finer raster ($0.05 \times 0.05 \times 0.05$ m). Differences between the infiltration patterns of Br and Brilliant Blue FCF were preserved by this interpolation method. The Br mass balance (cp. Section 3.1.1) that was calculated from the cokriged Br distribution confirmed the good representativity of the sampling scheme.

We conducted one solute displacement experiment at each land use treatment only. We acknowledge that the sampling design and the high experimental effort did not permit replicate experiments. This might raise the question about the representativity of our findings. The study of Schwen et al. (2013) showed negligible variations in soil texture and pedology along a transect across both experimental fields. Since the displacement experiments of this study were conducted in the center of each treatment, they were representative for the prevailing soil texture. However, variations in soil structure have been identified to be more relevant for solute displacement in this study. With respect to soil structure, a representative elementary volume (REV) should be covered by the experiment. To our knowledge, there exists no study that developed a method for deriving a REV for soil structure yet. This might be mainly due to a lack of suitable measurement methods. The typical occurrence of preferential flow pathways in the excavated profile sections varied on a 0.1 m-scale. With a volume of 0.40 m^3 , the excavated soil at each experimental site was on a larger scale and covered representatively the structural pore network. The

representativity of the experimental size was also confirmed by the high Br mass recovery (Fig. 11). Moreover, spatial ranges of representativity could be identified in our study that justify the approach for gaining detailed insights in flow and transport behavior under boundary conditions imposed and the scale of this investigation.

4.2. Proxies for soil structure characterization

Variations in soil structure have been ascribed to the bulk density in many studies. In agreement with Bormann and Klaassen (2008), we found a higher bulk density in the topsoil of the cropland compared to the grassland. The highest bulk densities were found in the subsoil of the cropland and were attributed to the existence of a plough pan. However, spatial variations of the bulk density agreed only moderately with the solute displacement patterns at the scale of this investigation. This was also revealed by the PCA (Fig. 10). As ρ_b is widely used as a proxy for soil structure – for instance in most pedotransfer functions (Vereecken et al., 2010) – this finding somehow questions if the bulk density contains relevant information about soil structure with respect to water and solute movement in soils and if bulk density should indeed be assigned prior importance in pedotransfer functions. This concern was also stressed by studies with focus on subsoil compaction (Håkansson and Lipiec, 2000; Schwen et al., 2011).

Our results also showed that the spatial patterns and variations of ρ_b and τ differed from each other – especially in the grassland (Figs. 7a–d and 8g–j). It can be concluded that they did not provide the same information about soil structure and its spatial variation. This finding was supported by a rather low correlation between both quantities of 0.233 (Table 3). It might be mainly due to the different size covered by the sample: The bulk density was derived from rather small core samples that did not contain information about the density conditions in the surrounding of the core. We acknowledge that larger sample sizes (e.g., 100 cm^3) might have better covered bulk density variations. Contrarily, the vane shear test delivered a mean shear strength that integrated over the surrounding soil volume in a radius of approximately 0.05 m. As stated by Horn (2004), the mechanical resistance of soil against shear forces mainly depends on the compaction, but also on the degree of aggregation and structure development. Thus, soil with a low density and a high proportion of structural pores inhibits a

lower mechanical shear strength than compacted soil under wheel tracks or in plough pans. This relation was confirmed by high τ values in the plough pan of the cropland and lower values in the topsoil of the grassland (Fig. 7). These results supported the statement of Horn (2004) who promoted the mechanical shear strength as an easily applicable measure for characterizing soil structural variations in space and time.

Soil aggregation and structure development is also strongly influenced by activities of burrowing soil fauna (e.g., earthworms) and plant roots. As reported by Bodner et al. (2014), growing and decaying plant roots are known to loosen the surrounding soil and produce bio-macropore networks with a high connectivity. Therefore, loosened and intensively-rooted soil is often associated with the occurrence of macropores and preferential flow areas (Fig. 5) (Kodešová et al., 2011; Nobles et al., 2010). The large negative correlation between the dye staining degree T and the mechanical shear strength τ (Table 3; Fig. 10) confirmed our assumption that soil structure development is linked to mechanical shear strength. We found an even better correlation between the Br displacement and τ (Table 3). This was very likely caused by considerable Br proportions that infiltrated along fine plant roots into the topsoil at the grassland (Fig. 11).

4.3. Land use effects on solute displacement

The results of our field experiment showed that both applied tracers – Br and Brilliant Blue FCF – followed different flow paths and their displacement differed between the tested land uses (grassland and cropland) (Fig. 4a–d, Video 1–4). Both tracers followed heterogeneous flow patterns as expected for a well-structured silt loam soil (Wang and Zhang, 2011). In the grassland, high proportions of the dye remained in or directly beneath preferential flow channels and macropores (Fig. 5). This macropore network consisted mainly of vertically orientated earthworm burrows and areas loosened by plant roots. Especially the soil directly below the surface (0–0.10 m depth) was intensively rooted by different grassland species. As a result of undisturbed biological activity for a long time, a network of well-connected, vertically orientated secondary pores (structural pores) had developed. This was also reported for the grassland sites studied by Bachmair et al. (2009). As reported by Weiler and Naef (2003), bypassing of water and solutes is enhanced by a low permeable soil matrix and a short interaction time. Our results indicated a high degree of macropore drainage in the grassland that was caused by a rather less permeable soil matrix throughout the analyzed profile depth. It indicated a rather low unsaturated hydraulic conductivity of the soil matrix.

The strong proportion of non-equilibrium flow through the macropore network and the bimodal pore-size distribution were also reflected by the Br displacement (Fig. 11). The largest proportion of the tracer was found directly below the soil surface (44%) and in the direct vicinity of the macropores. The tracer remaining close to the surface probably followed the grass roots that proliferated the soil. A considerable proportion of Br (17%) left the analyzed soil volume by preferential flow, either laterally or via the bottom. Compared to the cropland, the flow separation between both tracers was smaller as indicated by a somewhat higher correlation (Fig. 9).

Compared to the grassland, both tracers could better infiltrate into the soil matrix at the cropland site. This was reflected by a more homogeneous dye distribution with lower staining degrees compared to the grassland (Fig. 4, Videos 3 and 4). Also the Br was distributed more homogeneously and infiltrated deeper into the A horizon (Fig. 11). The flow process was characterized by a stronger lateral tracer movement from preferential flow paths to the adjacent soil. The flow separation between both tracers was slightly larger as reflected by a somewhat weaker correlation com-

pared to the grassland (Fig. 9). This indicated a larger proportion of medium-sized pores and a larger unsaturated hydraulic conductivity of the soil matrix. Our finding agreed with Bachmair et al. (2009) who characterized the pore network of their studied cropland sites as densely spaced, tortuous interaggregate voids. The mass balance showed that the largest proportion of Br (73%) was found in the A horizon. The old plough pan might have reduced the number of vertically continuous macropores and caused a longer time and amount of dye retained in the surface soil and thus more time to infiltrate into the soil matrix. This behaviour was also reported by Kodešová et al. (2011) and Sander and Gerke (2007). However, some macropores had also developed in the B and C horizon. As a consequence, 20% of the applied Br was found in these horizons (Fig. 11). Losses due to lateral or deep drainage were smaller (7%) than at the grassland site.

Both experimental sites of this study have been transferred to their current land use some 10 years ago. They were once part of one tobacco field and have a similar soil texture. Moreover, Horn (2004) reported an increased aggregate formation and abundance of more continuous macropores under no-till cropland compared to conventionally managed cropland. Thus, the two land use systems of our study might be more similar in terms of physical and hydrological properties than it might have been expected if the cropland had been conventionally managed with annual moldboard ploughing. For instance, treatments analyzed in the study of Bachmair et al. (2009) were longer established. With respect to the grassland, we therefore acknowledge that the findings of this study may not represent the effects that can be expected after several decades of grassland but rather the conditions some 10 years after land-use change (transversion into grassland).

As stated above, conservative tracers such as Br could be used as proxy for fertilizer application, while Brilliant Blue FCF indicates flow paths of larger organic molecules such as pesticides. Our results indicated that most of the applied tracers remained in the rooting zone at both land uses. We can conclude that most of the applied solutes were found at a depth where they could be either taken up by plants or become decomposed by microorganisms. The mass balance revealed that 22% and 16% of the total applied Br were leached below the root zone (deeper than 0.50 m) at the grassland and cropland, respectively (Fig. 11). Taking into account the rather extreme ponding head infiltration conditions, the risk of fertilizer or pollutant leaching into the karst groundwater is relatively small at this site. The leaching risk might be slightly larger at the grassland site due to a larger abundance of vertically orientated macropores.

5. Conclusion

In the present study, a field experiment was conducted to assess the impacts of land use (grassland vs. no-till cropland) on profile-scale solute displacement in a silt loam soil. Since comprehensive studies on solute displacement and the detection of flow paths and phenomena in field soils require a detailed picture about the distribution of the applied tracer and physical soil properties, some new measurement and data analysis methods were tested in the course of this study. Wavelength dispersive X-ray fluorescence (WD-XRF) proved well for the synchronous determination of Br concentrations and the relative SiO₂ signal intensity. The latter performed well as a proxy for soil textural variations in the analyzed soil. This study was the first application of the XRF method within a field solute displacement experiment. It showed a way how increased sample numbers can be analyzed efficiently – both with respect to sample preparation time and analytical expenses. More research is required to proof the correlation between major element signal intensities and particle-size distribution for other textural classes.

A new sampling scheme was tested where all observations except the dye staining pattern were determined as point measurements within regular $0.10 \times 0.10 \times 0.10$ m rasters. Detailed Br distributions were derived by cokriging on the basis of the dye staining patterns. The Br mass balance confirmed the representativity of this procedure.

Spatial correlations and the results of a PCA revealed that the actual land use was mainly reflected by variations in the bulk density, whereas spatial variations of the solute displacement were more correlated with small-scale variations in soil texture, water retention, and mechanical shear strength. The latter was found to strongly reflect soil structural variations that are relevant for solute transport processes in the vadose zone. This finding could stimulate future investigations to improve pedotransfer functions for better describing hydrologically relevant soil properties.

Although both experimental sites were once part of the same land use, Br displacement differed between the actual land uses. We detected a higher proportion of non-equilibrium flow through vertically orientated macropores at the grassland site. The higher degree of macropore drainage was also caused by a less permeable soil matrix. The continuous biological activity had two major implications for solute transport: Earthworm burrows were the main reason for the vertically orientated, highly conductive macropores. Since the site had been transferred into grassland some decades ago, it is also noticeable that biological structure development had withdrawn any compaction in the subsoil. Plant root growth and decay had formed a patchy network of loosened, densely rooted areas close to the surface. Here, large proportions of the Br could infiltrate into the shallow soil matrix.

The spatial Br distribution in the cropland was more homogeneous, with the majority found in the A-horizon. The soil structure of the topsoil was characterized by a coarser matrix with well-connected inter-aggregate pores. This effect resulted in a stronger lateral tracer movement from preferential flow pathways to the adjacent soil. Soil structure development at the cropland site was mainly controlled by agricultural operations such as tillage in former times and seedbed preparation by chisel application. These operations were also the reason for the development of a plough pan that restricted solute displacement to deeper soil or to ground-water bodies. Although ponding infiltration was applied, the leaching risk for both applied solutes – Br and Brilliant Blue FCF – was rather small.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jhydrol.2014.09.028>.

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